

TRADE ASSOCIATION NETWORKS AND GLOBAL NTM TRADE POLICY FORMATION

Community Data Lab Project Report

World Bank

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Introduction and Research Objective

The purpose of this project is to investigate the relationship between trade association density and the use of non-tariff measures (NTMs) across industries and countries over time. Non-tariff measures have become an increasingly important component of global trade policy, particularly as tariff barriers have declined due to commitments under international trade agreements. As tariffs have fallen, governments may substitute toward alternative regulatory mechanisms such as technical barriers, licensing requirements, or sanitary standards that restrict trade indirectly.

The central research question guiding this project is:

Do industries with greater trade association density experience higher levels of non-tariff trade protection across countries and over time?

This research is motivated by the political economy literature on trade policy formation. The “Protection for Sale” framework suggests that organized interest groups can influence trade policy outcomes through lobbying activities. Trade associations represent a coordinated mechanism through which industries may advocate for regulatory protections that benefit domestic producers. If trade associations successfully influence policy formation, industries with stronger organizational networks may exhibit higher levels of non-tariff protection.

Progress Report 1 focused primarily on data construction and pipeline development required to integrate multiple datasets recorded at different levels of classification. The current report builds on that work by conducting initial exploratory analyses of the constructed dataset and outlining the empirical strategy that will be implemented in the final stage of the project.

Data Construction Summary

The analysis combines several datasets that operate at different classification levels:

Non-Tariff Measure Data (WITS)

The NTM dataset from the World Integrated Trade Solution (WITS) records regulatory measures applied to traded goods. Each observation identifies the country implementing the measure, the product code affected, the type of NTM, and the year of implementation.

Trade Association Data

Trade association information is obtained from the World Guide to Trade Associations. These data provide counts of trade associations operating within specific industries across countries.

Industry Characteristics (INDSTAT)

Industry-level economic variables such as employment, output, value added, wages, and number of establishments are drawn from the UNIDO INDSTAT database.

A major challenge in constructing the dataset arises from the fact that NTM data are recorded at the HS product level, while trade association and industry statistics are recorded at the ISIC industry level. To resolve this mismatch, the following data pipeline was implemented:

- Raw NTM data were cleaned and standardized, including formatting country names and product codes.
- NTM observations were collapsed to the country–year–HS product level to count the number of measures affecting each product.
- HS codes were mapped to ISIC Revision 3 industry codes using official concordance tables.
- The data were then re-collapsed to the country–year–industry level.
- Finally, the resulting dataset was merged with trade association data and INDSTAT industry characteristics.

The resulting dataset is a country–year–industry panel, where each row represents a specific industry within a country and year. Columns include counts of NTMs by type as well as industry-level economic variables.

Initial Exploratory Analysis

With the preliminary dataset constructed, the next step was to conduct exploratory analyses to understand the structure of the data and identify patterns that may inform the empirical model.

Distribution of Non-Tariff Measures

Initial descriptive statistics reveal that the distribution of NTMs varies substantially across industries and countries. Some industries exhibit very high counts of regulatory measures, while others have relatively few recorded NTMs. This suggests that trade protection through non-tariff measures may be concentrated within particular sectors.

The dataset also records NTMs by category. These include measures such as sanitary and phytosanitary regulations, technical barriers to trade, licensing requirements, and export-related restrictions. The variation across NTM categories provides an opportunity to construct multiple measures of NTM intensity for use in regression analysis.

Industry-Level Variation

Preliminary inspection of the industry-level panel indicates substantial heterogeneity in NTM counts across ISIC industries. Certain manufacturing industries appear to have particularly high counts of NTMs. This pattern is consistent with the expectation that industries facing international competition may rely more heavily on regulatory protections.

Because the dataset also includes industry characteristics such as employment and output, it will be possible to control for industry size and economic importance when estimating the empirical model.

Country-Level Trends

Initial examination of the data suggests that NTM implementation has increased over time in many countries. This pattern aligns with broader global trends in trade policy, where governments increasingly rely on regulatory measures rather than tariffs to influence trade flows.

These preliminary patterns support the motivation for the project and suggest that the dataset contains meaningful variation that can be used to test the research hypothesis.

Data Cleaning and Processing

Before conducting the analysis, the raw NTM dataset required substantial cleaning and preprocessing. The original dataset contains millions of observations recorded at the product level using Harmonized System (HS) codes. These observations include inconsistencies in country names, product codes, and variable formats.

To address these issues, a data processing pipeline was implemented using Python and DuckDB, which allows efficient processing of large datasets without loading the entire dataset into memory.

The main cleaning steps included:

- Standardizing country names and converting them to uppercase.
- Extracting the year variable from the policy start date.
- Cleaning product codes by removing non-numeric characters.
- Removing observations with missing country or product codes.

After cleaning, the dataset was aggregated to the country–year–HS code level, where the number of NTMs affecting each product was counted. This aggregation reduces duplication and prepares the dataset for mapping to industry-level classifications.

This cleaning pipeline ensures that the dataset used for analysis is consistent and reproducible.

Empirical Strategy

The empirical analysis will estimate the relationship between trade association density and NTM intensity using a panel regression framework.

The baseline specification is:

$$NTM_{c,i,t} = \beta \cdot TA_Density_{c,i,t} + X_{c,i,t} + \alpha_{c,i} + \gamma_t + \epsilon_{c,i,t}$$

Where:

- $NTM_{c,i,t}$ represents NTM intensity in country c , industry i , and year t .
- $TA_Density_{c,i,t}$ measures trade association density within that industry.
- $X_{c,i,t}$ represents industry-level control variables from INDSTAT.
- $\alpha_{c,i}$ represents country fixed effects.
- γ_t represents year fixed effects.

The primary coefficient of interest is β , which captures the relationship between trade association density and NTM intensity.

Several alternative measures of NTM intensity will be tested to ensure robustness. These include:

- Total number of NTMs affecting an industry
- Binary indicator for the presence of any NTM
- Counts of specific NTM categories
- Trade-weighted NTM coverage (if trade data are incorporated)

Standard errors will be clustered at the country level to account for potential correlations in policy implementation within countries.

Example Code and Result

The code I have below is for the Canada data. The code cleans and transforms the raw Canada WITS NTM dataset into an industry–year panel dataset used for analysis. It first loads the raw NTM data and merges it with the H4 HS–ISIC concordance table to map product-level HS codes to ISIC Revision 3 industry codes. Before merging, the code standardizes column names and converts product codes to numeric format to ensure the merge works correctly.

Next, the code extracts the year from the NTM start date and creates an NTM category variable (`ntm_type`) using the first letter of the NTM code. Unnecessary columns that are no longer needed for analysis are removed to simplify the dataset.

The data are then aggregated by industry, NTM type, and year to count the number of measures affecting each industry. After aggregation, the dataset is reshaped into a wide format, where each NTM category becomes its own column (e.g., `countA`, `countB`, `countE`). The final dataset contains one row per industry–year observation. Finally, the cleaned dataset is saved in both `.dta` and `.csv` formats for later merging with trade association and industry-level data.

```
import os
import pandas as pd

# -----
# Canada (H4 -> merge on hs2012productcode using h4i3cw.dta)
# -----
```

```

can_csv = r"Z:\Data\Country NTM Downloads (Raw WITS
Data)\Cs\Canada\DataJobID-3047732_3047732_Canada.csv"
h4i3cw_dta = r"Z:\Data\WITS Crosswalks\h4i3cw.dta"
can_dir = r"Z:\Data\Country NTM Downloads (Raw WITS Data)\Cs\Canada"

df = pd.read_csv(can_csv)

df.columns = df.columns.str.strip().str.lower()

if "year" in df.columns:
    df.rename(columns={"year": "reported_year"}, inplace=True)

df.rename(columns={"productcode": "hs2012productcode"}, inplace=True)
df["hs2012productcode"] = pd.to_numeric(df["hs2012productcode"],
errors="coerce").astype("Int64")

cw = pd.read_stata(h4i3cw_dta)
cw.columns = cw.columns.str.strip().str.lower()
cw["hs2012productcode"] = pd.to_numeric(cw["hs2012productcode"],
errors="coerce").astype("Int64")

merged = df.merge(cw, how="left", on="hs2012productcode", indicator=True)

merged = merged[merged["_merge"] != "left_only"].copy()
merged.drop(columns=["_merge"], inplace=True)

merged.rename(columns={"isicrevision3productcode": "isic_rev3"},
inplace=True)

merged["year"] = pd.to_numeric(merged["startdate"].astype(str).str[:4],
errors="coerce")

merged["ntm_type"] = merged["ntmcode"].astype(str).str[0]

drop_cols = [
    "ntmnomenclature",
    "ntmcode",
    "partialcoverage",
    "reporter_iso_n",

```

```

        "year_productclassification",
        "startdate",
        "enddate",
        "partner",
        "nomencode",
        "reportername",
        "reported_year",
        "hs2012productdescription",
        "isicrevision3productdescription",
        "productcode",
    ]

merged.drop(columns=[c for c in drop_cols if c in merged.columns],
            inplace=True)

merged.loc[merged["year"] < 1990, "year"] = 1990

merged = merged[merged["isic_rev3"].notna()].copy()
merged["count"] = 1

collapsed = (
    merged.groupby(["isic_rev3", "ntm_type", "year"],
                  as_index=False) ["count"]
        .sum()
)

os.makedirs(can_dir, exist_ok=True)
os.chdir(can_dir)

years = sorted(collapsed["year"].dropna().astype(int).unique().tolist())
if len(years) == 0:
    raise ValueError("No years found after processing. Check merge +
startdate parsing.")

for y in years:
    sub = collapsed[collapsed["year"] == y]

    wide = sub.pivot_table(
        index="isic_rev3",
        columns="ntm_type",
        values="count",
    )

```



```

        aggfunc="sum"
    ).reset_index()

    wide.columns = ["isic_rev3"] + [f"count{c}" for c in wide.columns[1:]]

    wide["year"] = y

    wide.to_stata(f"temp_{y}.dta", write_index=False)

base = pd.read_stata(f"temp_{years[0]}.dta")
for y in years[1:]:
    base = pd.concat([base, pd.read_stata(f"temp_{y}.dta")],
                    ignore_index=True)

base["country"] = "canada"

dta_path = os.path.join(can_dir, "canada_ntms.dta")
csv_path = os.path.join(can_dir, "canada_ntms.csv")

base.to_stata(dta_path, write_index=False)
base.to_csv(csv_path, index=False)

print("Saved final files:")
print(dta_path)
print(csv_path)

```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	NTM Nomenclature	ReporterName	Year	NTMCode	NomenCode	ProductCode	PartialCoverage	Partner	Reporter_ISO_Year	ProductClassification	StartDate	EndDate			
2	M4	Australia	2016	A14	H4	1012100	0	WLD	36		2015	7/7/1998	4/1/2019		
3	M4	Australia	2016	A14	H4	1012900	0	WLD	36		2015	7/7/1998	4/1/2019		
4	M4	Australia	2016	A14	H4	1013000	0	WLD	36		2015	7/7/1998	4/1/2019		
5	M4	Australia	2016	A14	H4	1019000	0	WLD	36		2015	7/7/1998	4/1/2019		
6	M4	Australia	2016	A14	H4	1022100	0	WLD	36		2015	7/7/1998	4/1/2019		
7	M4	Australia	2016	A14	H4	1022900	0	WLD	36		2015	7/7/1998	4/1/2019		
8	M4	Australia	2016	A14	H4	1023100	0	WLD	36		2015	7/7/1998	4/1/2019		
9	M4	Australia	2016	A14	H4	1023900	0	WLD	36		2015	7/7/1998	4/1/2019		
10	M4	Australia	2016	A14	H4	1029000	0	WLD	36		2015	7/7/1998	4/1/2019		
11	M4	Australia	2016	A14	H4	1031000	0	WLD	36		2015	7/7/1998	4/1/2019		
12	M4	Australia	2016	A14	H4	1039100	0	WLD	36		2015	7/7/1998	4/1/2019		
13	M4	Australia	2016	A14	H4	1039200	0	WLD	36		2015	7/7/1998	4/1/2019		
14	M4	Australia	2016	A14	H4	1041000	0	WLD	36		2015	7/7/1998	4/1/2019		
15	M4	Australia	2016	A14	H4	1042000	0	WLD	36		2015	7/7/1998	4/1/2019		
16	M4	Australia	2016	A14	H4	1051100	0	WLD	36		2015	7/7/1998	4/1/2019		

↓

	A	B	C	D	E	F	G	H	I	J	K	L
1	isic_rev4	countA	countB	countE	countP	year	countF	countC	countN	country		
2	111	0	5	2	125	1990				australia		
3	112	0	5	2	125	1990				australia		
4	113	0	5	3	238	1990				australia		
5	114	0	5	2	125	1990				australia		
6	115	0	5	2	125	1990				australia		
7	116	0	5	2	125	1990				australia		
8	119	0	5	3	238	1990				australia		
9	121	0	0	10	233	1990				australia		
10	122	0	0	10	233	1990				australia		
11	123	0	0	10	233	1990				australia		
12	124	0	0	10	233	1990				australia		
13	125	0	0	11	346	1990				australia		